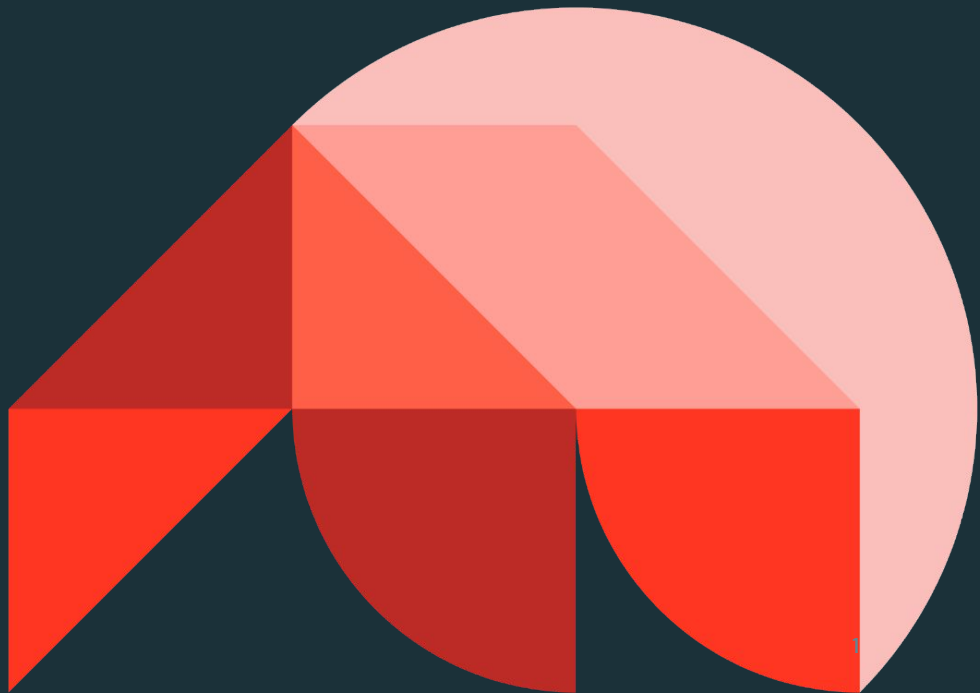




Databricks Unity Catalog

July 2024



Agenda

1. Overview
2. Unity Catalog Upgrade
3. Example



Unity Catalog Overview



Databricks Unity Catalog

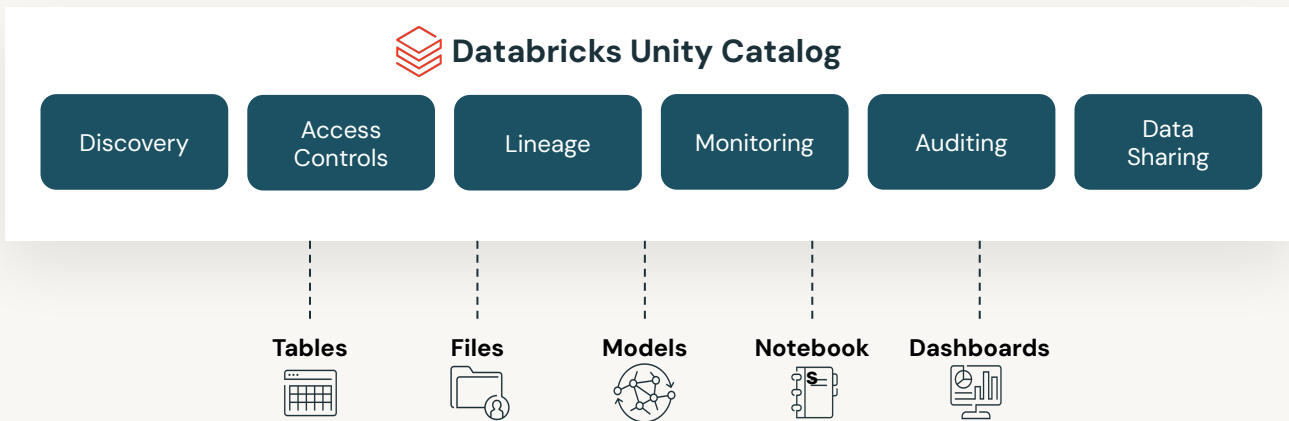
Unified governance for data and AI

Unified visibility into data and AI

Single permission model for data and AI

AI-powered monitoring and observability

Open data sharing



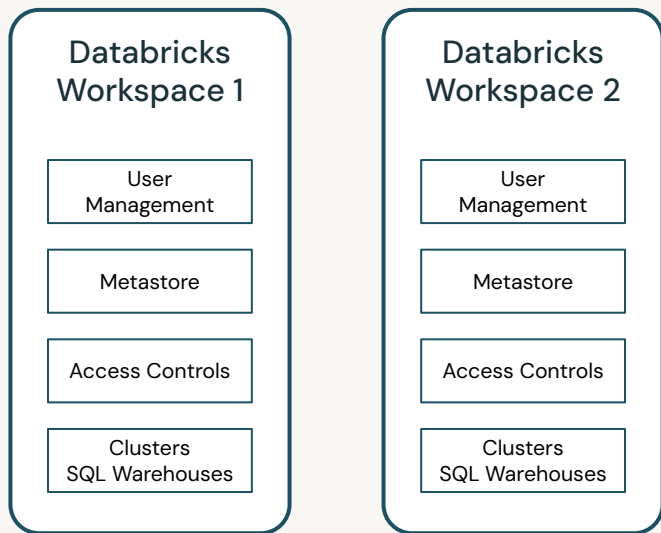
Unity Catalog Unlocks Full Potential of your Lakehouse	w/ UC 	w/ Hive Metastore 
Governance across workspaces (Access controls, metastore, Identity management)	✓	✗ (Governance at workspace level)
Automated table and column level lineage in real-time	✓	✗
Row Filters and Column Masking (use standard SQL functions for fine-grained access controls)	✓	✗
Volumes (Govern non tabular data/arbitrary files – image, video, PDF etc.)	✓	✗
Lakehouse Federation (Discover, govern and query external databases, data warehouses, Hive Metastore, AWS Glue)	✓	✗
Lakehouse Monitoring (Monitor quality of data and ML models with auto generated alerts and dashboards)	✓	✗
Delta Sharing (Share files, tables, ML models, notebooks across clouds, regions and data platforms)	✓	✗
Databricks Marketplace	✓	✗
Databricks Cleanroom (Collaboration on data in a privacy-preserving manner)	✓	✗
Databricks Data Room (aka Genie)	✓	✗
Databricks Assistant / LakehouseIQ (Knowledge engine on the lakehouse to get insights in natural language)	✓	✗ (Degraded experience)
Governance for AI (manage and govern ML Models, Feature store)	✓	✗
System Tables (end to end observability for Billing, Auditing, Lineage, Marketplace analytics)	✓	✗
Lakehouse Apps	✓	✗
Databricks Connect V2	✓	✗
Materialized Views	✓	✗
AI Powered Predictive optimization (optimizes your query plans and data layout for peak performance, intelligently balancing read vs. write performance)	✓	✗
Streaming Tables	✓	✗
HMS Interface (Query data registered in UC using other data platforms such as Amazon Athena, Presto, Trino,EMR)	✓	✓
Vector Search for indexing	✓	✗
Serverless workflows	✓	✗



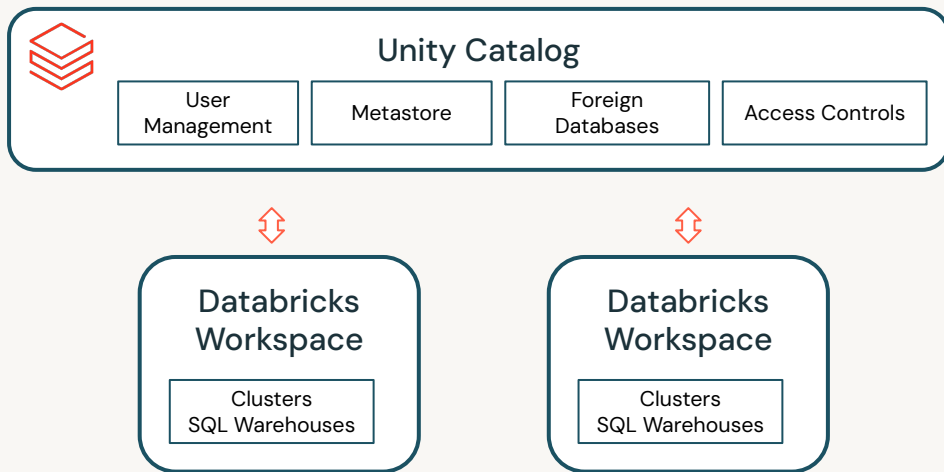
All your metadata, in one place

One metadata layer across file and database sources **superpowers** governance

Without Unity Catalog

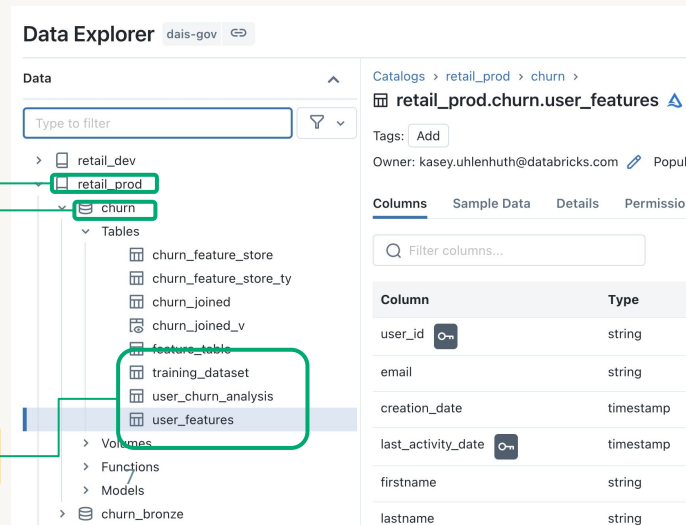
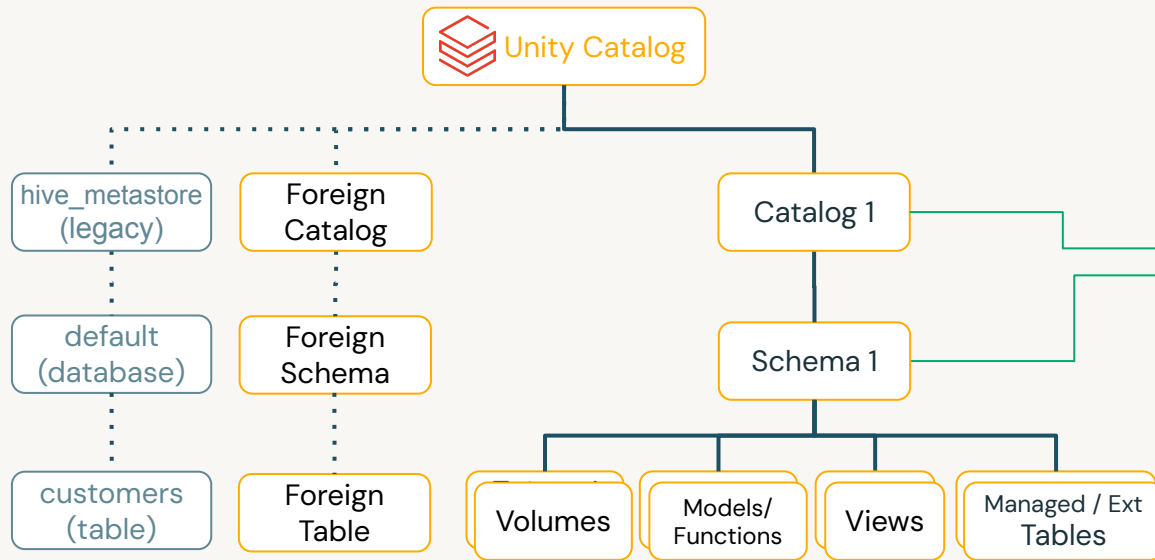


With Unity Catalog



Governed namespace across file and database sources

Access legacy metastore and foreign databases powered by Lakehouse Federation



```
SELECT * FROM main.paul.red_wine; -- <catalog>.<database>.<table>
```

```
SELECT * FROM hive_metastore.default.customers;
```

```
SELECT * FROM snowflake_warehouse.some_schema.some_table;
```



Centralized Access Controls

Centrally grant and manage access permissions across workloads

Using ANSI SQL DCL

```
GRANT <privilege> ON <securable_type>  
<securable_name> TO <principal>
```

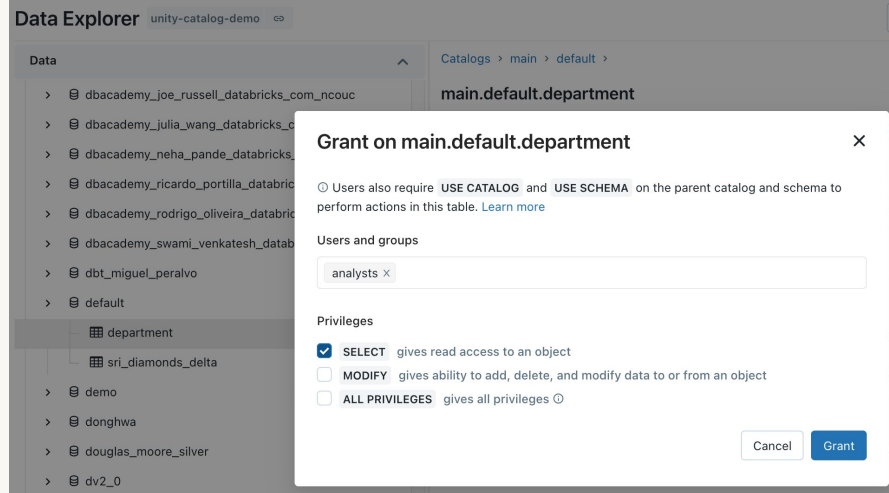
```
GRANT SELECT ON iot.events TO engineers
```

Choose
permission level

'Table'= collection of
files in S3/ADLS

Sync groups from
your identity
provider

Using UI



Row Level Security and Column Level Masking

Provide differential fine grained access to datasets

Only show specific rows

```
CREATE FUNCTION <name> ( <parameter_name >  
<parameter_type> .. )  
RETURN {filter clause whose output must be a boolean}  
  
CREATE FUNCTION us_filter(region STRING)  
RETURN IF(IS_MEMBER('admin'), true, region="US");  
  
ALTER TABLE sales SET ROW FILTER us_filter ON region;
```

Test for group membership

Assign reusable filter to table

Specify filter predicates

Mask or redact sensitive columns

```
CREATE FUNCTION <name> (<parameter_name>,  
<parameter_type>, [, <column>...])  
RETURN {expression with the same type as the first  
parameter}  
  
CREATE FUNCTION ssn_mask(ssn STRING)  
RETURN IF(IS_MEMBER('admin'), ssn, "*****");  
  
ALTER TABLE users ALTER COLUMN table_ssn SET MASK  
ssn_mask;
```

Test for group membership

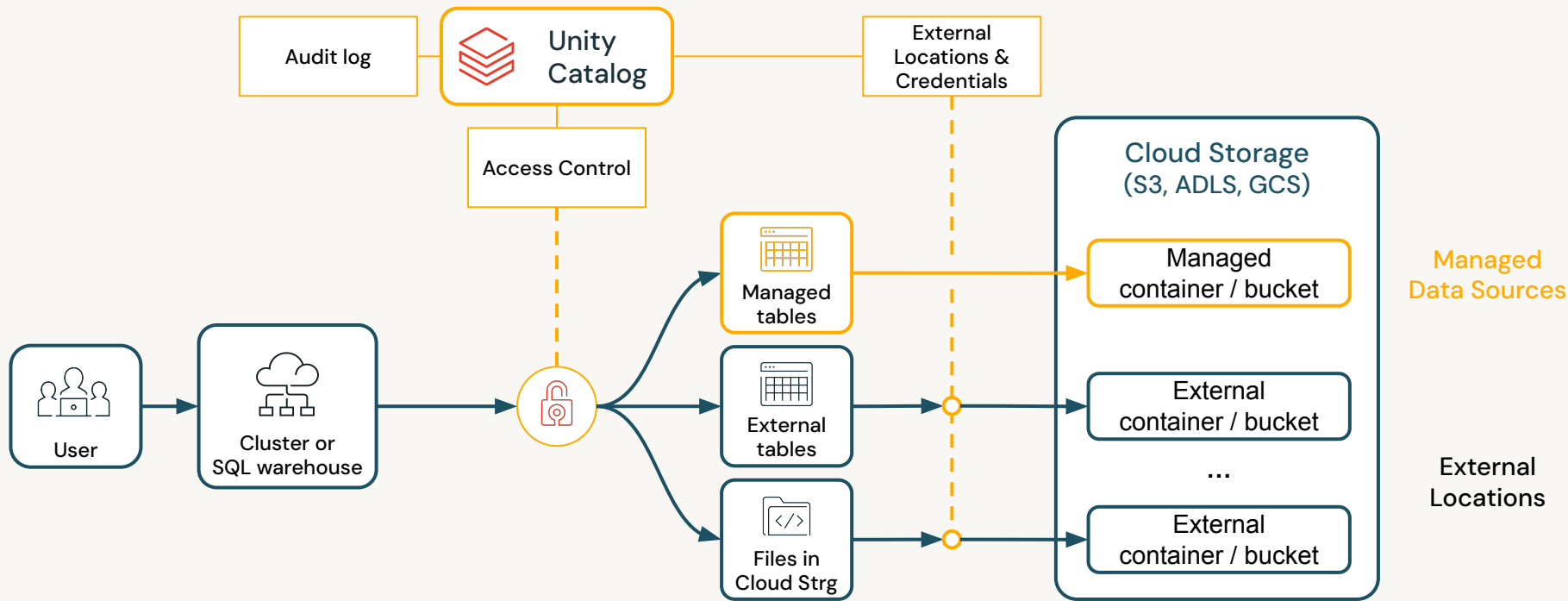
Assign reusable mask to column

Specify mask or function to mask

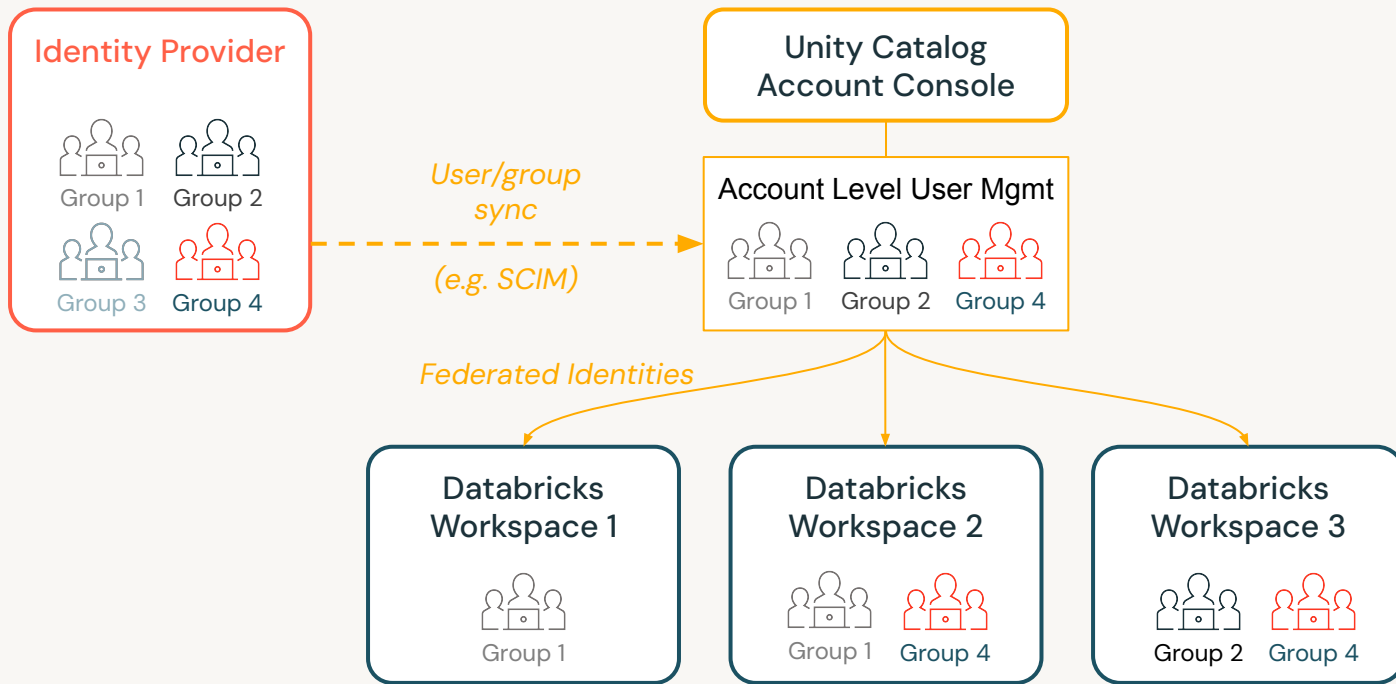


Managed Data Sources & External Locations

Simplify data access management across clouds



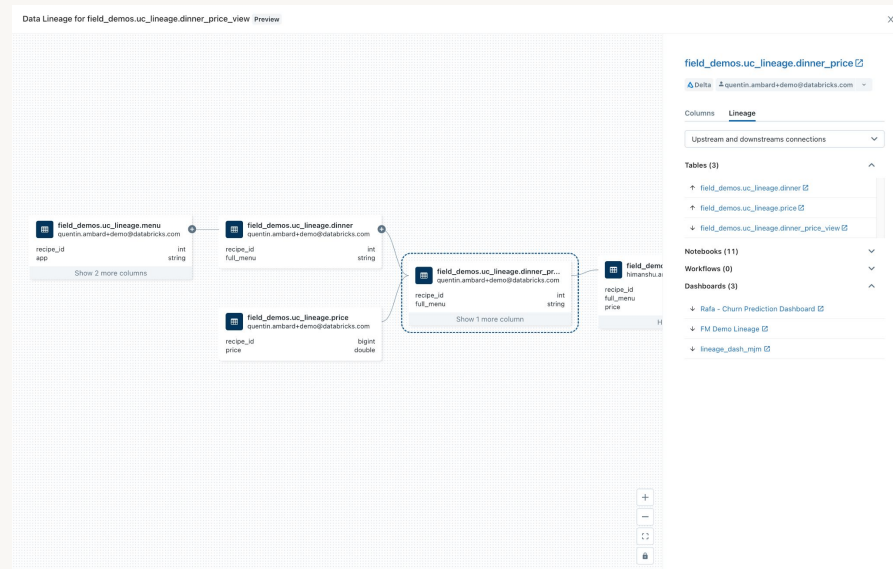
User management with Unity Catalog



Automated lineage for all workloads

End-to-end visibility into how data flows and consumed in your organization

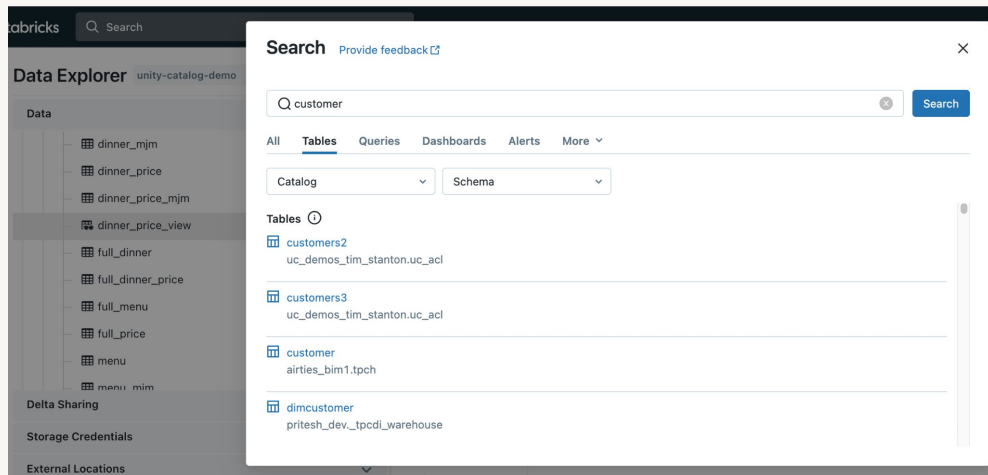
- Auto-capture runtime data lineage on a Databricks cluster or SQL warehouse
- Track lineage down to the table and column level
- Leverage common permission model from Unity Catalog
- Lineage across tables, dashboards, workflows, notebooks, feature tables, files, and DLT



Built-in search and discovery

Accelerate time to value with low latency data discovery

- UI to search for data assets stored in Unity Catalog
- Unified UI across DSML + DBSQL
- Leverage common permission model from Unity Catalog
- Apply semantic tags to data and search across tags



Discovery Tags

Semantic layer for your lakehouse

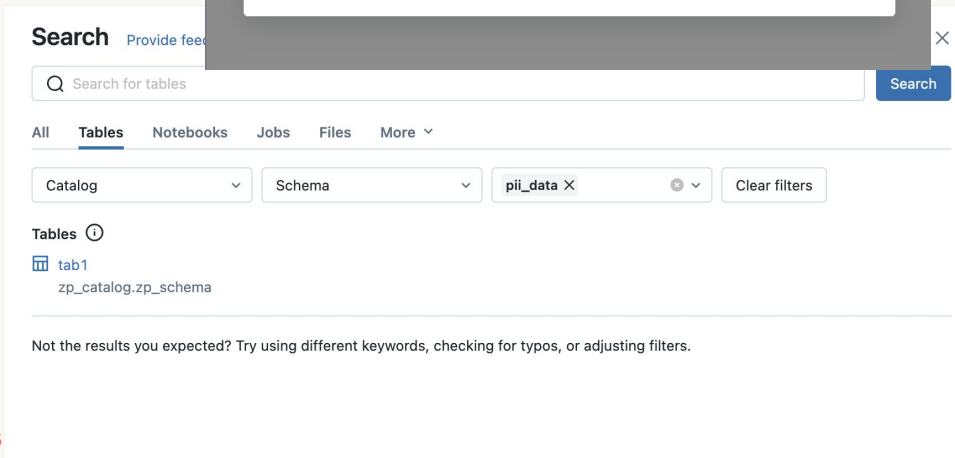
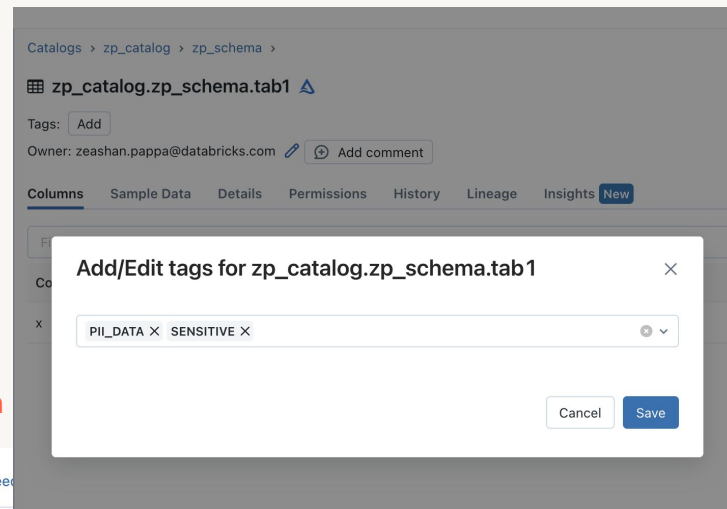
Problem

Searching for data assets in business terms or generally agreed upon taxonomies usually requires additional catalog tools.

Solution

- Discovery Tags allow you to tag Column, Table, Schema, Catalog objects in UC
- **Integrated search mechanism in UC allows you to search for objects by tag.**

Tag your data



Search based on tags



Volumes in Unity Catalog

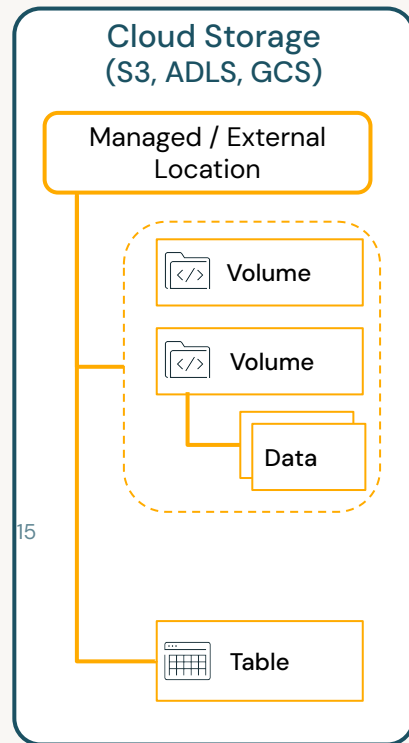
Access, store, organize and process files with Unity Catalog governance

- Volumes can be accessed by POSIX commands

```
dbutils.fs.ls("s3://my_external_location/Volumes/volume123")
```

```
ls /Volumes/volume123
```

- Volumes are created under Managed or External Locations and show up in UC Lineage
- Volumes add governance over non-tabular data sets
 - Unstructured data, e.g., image, audio, video, or PDF files, used for ML
 - Semi-structured training, validation, test data sets, used in ML model training
 - Raw data files used for ad-hoc or early stage data exploration, or saved outputs
 - Library or config files used across workspaces
 - Operational data, e.g., logging or checkpointing output files
- Tables are registered in Managed / External Locations, not in Volumes





Unity Catalog Upgrade Steps



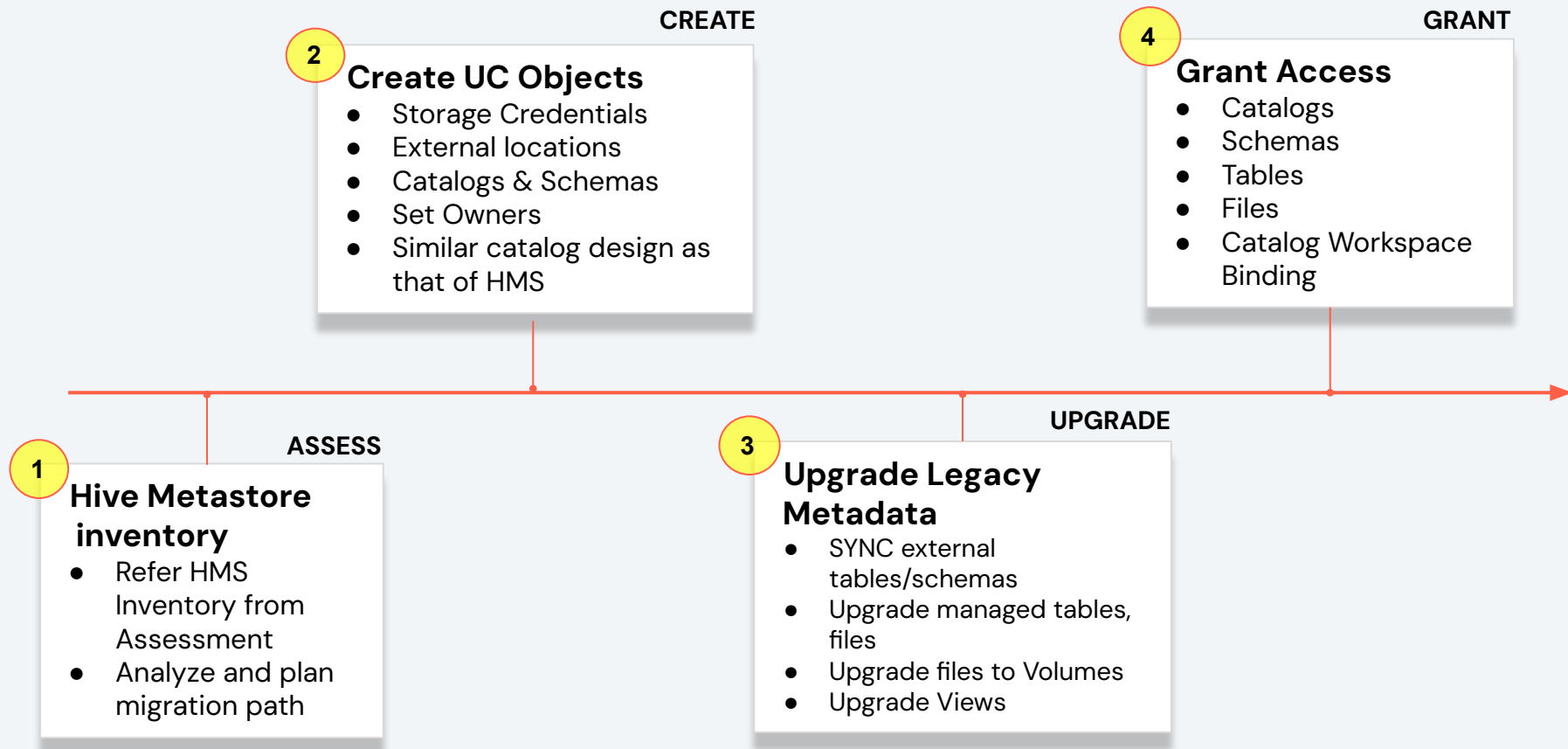
Unity Catalog Upgrade Steps

1. Finalize UC Design – Catalogs, Workspaces, Account Groups
2. Generate UCX Assessment, Estimate Effort
3. Create Metastore and Assign to Workspace
4. Migrate Workspace Local Groups to corresponding Account Level groups
5. Create storage credentials, external locations, catalogs
6. SYNC EXTERNAL Tables and replicate Managed tables
7. Grant access to catalogs, schemas, tables, files
8. Migrate workloads – Create Clusters , Warehouses and Jobs, Update Notebooks and downstream tools
9. Decommissioning – old pipelines, clusters, HMS, mounts

Deep Dive - Upgrading Data

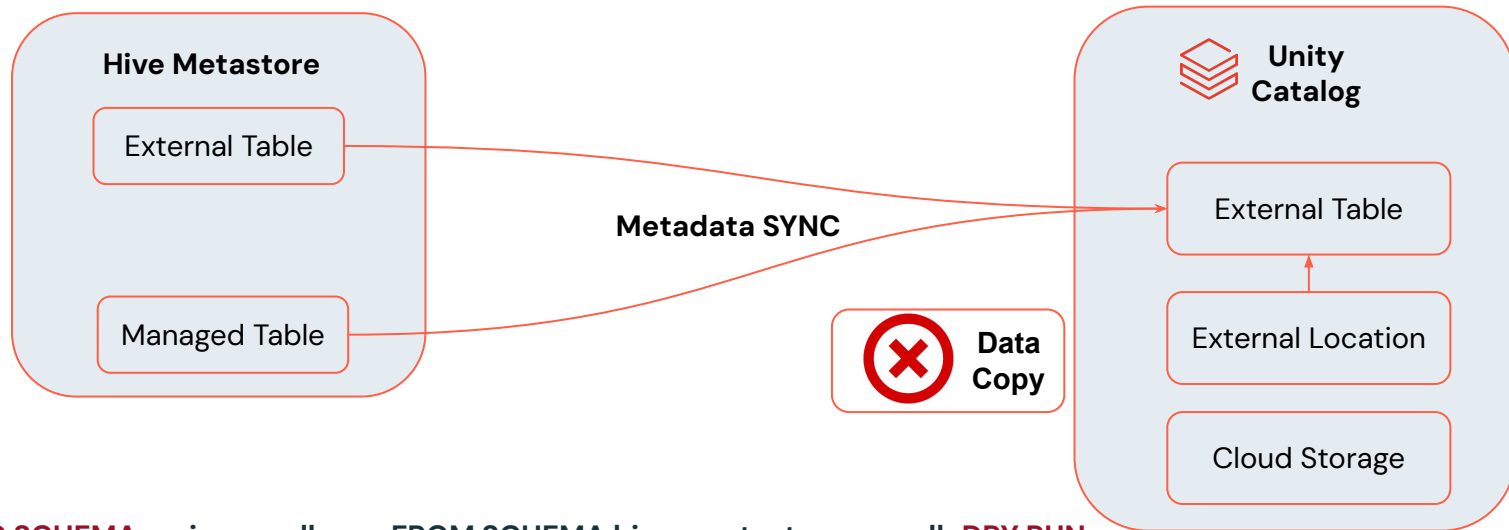
Upgrade Data to Unity Catalog

Steps to consider for upgrading Data



Upgrade Data to Unity Catalog

SYNC Means **no data copy**



SYNC SCHEMA main.my_db_uc FROM SCHEMA hive_metastore.my_db **DRY RUN**

SYNC TABLE main.my_db_uc.my_tbl FROM hive_metastore.my_db.my_tbl

Data upgrade scenarios

Upgrade Methodology Matrix

HMS Table Type	Target UC Table Type	Data Storage Location	Methodology
External or Managed	External or Managed	DBFS Root	CTAS/Deep Clone
External or Managed	External	DBFS Mounted Storage or Cloud Storage	SYNC
External or Managed	Managed	DBFS Mounted Storage or Cloud Storage	CTAS/Deep Clone
Hive SerDe Interface			
External or Managed	External or Managed	NA	CTAS/Deep Clone

Deep Dive Upgrading Compute

UC Clusters

What's That? And why should I care?

UC clusters = **can access data in UC**

Requirements:

1. UC is enabled for workspace
2. Cluster in 'Shared' or 'Single User' access mode
3. No legacy features selected
(credentials passthrough etc)

All other clusters cannot access data in UC

Summary

1-2 Workers	30.5-61 GB Memory 4-8 Cores
1 Driver	30.5 GB Memory, 4 Cores
Runtime	13.3.x-scala2.12

Unity Catalog

Photon

i3.xlarge

4-6 DBU/h

Upgrade Clusters to Unity Catalog

SQL Warehouse

SQL Warehouse Clusters

- Under Advanced options
- Toggle Unity Catalog option
- Use workspace level default catalog configuration to work for both DBR clusters and SQL Warehouses.

Advanced options ▾

Tags ⓘ

Key	Value
-----	-------

Unity Catalog ☒

Channel ⓘ ☒ Current ☐ Preview

Self learning Resources



Pre-workshop Learning Resources

- [UC best practices document](#) (creat shared): Refer to page 40 in the PDF for "Job upgrade guidance"
- [Unity catalog essentials](#) : End-to-end course from Databricks Academy, covering UC overview, compute resources and UC, data access control in UC, UC patterns and best practices. You can pick and choose specific modules within the course as well
- [Unity Catalog – Databricks official documentation](#)



Other Learning Resources

- If you are newer to Databricks and want to learn more about the platform/unity catalog, please check out a few other relevant free learning videos from Databricks Academy below.
 - [Volumes in unity catalog](#)
 - [Creating external and managed tables in unity catalog](#)
 - [Creating a cluster in unity catalog](#)
 - [Data lineage with unity catalog](#)
 - [Assigning a metastore to a workspace in unity catalog](#)
 - [Assigning metastore administrators in unity catalog](#)
 - [Creating and deleting metastores in unity catalog](#)
 - If you want to learn more about the features unlocked by Unity Catalog within the Databricks Platform: [Lakehouse Federation](#), [Lakehouse Monitoring](#), [Delta Sharing](#), [Databricks Marketplace](#), [Databricks Cleanroom](#), [Databricks Genie Spaces\(preview\)](#), [System Tables](#), [Lakehouse Apps](#), [Databricks Connect v2](#), [Materialized Views](#), [AI powered Predictive Optimization](#), [Streaming Tables](#), [Vector Search for indexing](#), [Serverless workflows](#)

Note: Everyone from Nestle will have access to [Databricks Academy](#), please reach out if you have any issues registering with your nestle email/logging in.



Q&A

